

Final project - Thermal Comfort Prediction

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For the pre-process, the first step was to merge the metadata set that includes climate and cooling type. First, we deleted the features which have a greater proportion of missing values such as blind_curtain, window, door, fan. We also drop the tr, tg because of missing values and are similar to the ta features. For features that are unrelated to other features of the environment (age, clothing insulation, metabolic rate, height, weight), we would simply take the average. Other features such as gender, are difficult to determine a value based on a feature and we did not want to simply take the average so we decided to have an even distribution data for that feature (fill the NA randomly by 'male' or 'female'). When it comes to features related to temperature and humidity, this is where we can relate them with the metadata climate and cooling type. So to calculate the outdoor air we grouped by climate and season and took the average. So for all nan values in t_out, depending on its climate and current season, it would take the average of that group computed across all non-nan data. This was also done for rh_out. For the velocity, we simply grouped by cooling type and computed the mean and filled out the missing values depending on which group they belong to. Typically naturally ventilated cooling types will have higher velocity compared to air conditioned cooling types. Finally to handle text data, we used a OneHotEncoder for gender, season, climate and cooling type. Which will create a separate column for each value type for that feature and assign a 1 or 0. We create a function to handle each process (data imputation, one hot encoding).

We decided to select the XGBoost model because we have a classification problem and we want a model that can handle continuous, categorical features and missing data. XGBoost stands out for its speed and performance when the dataset is large. The model can be obtained by using the xgboost package and we also import some functions from the skopt package to compute the accuracy score and to select the optimal hyperparameters via Bayes search. The hyperparameters we are looking for are the subsample, the minimum child weight, the learning rate, the maximum depth, the columns sample by tree and by level, the regularization parameter (gamma) and the L2 regularization term. The classification benefits from this non-linear model because of the use of continuous and discrete features and some features are related to each other because they are temperature and humidity measures. Because of the complexity of the model, the hyperparameters computation search time is about 10 or 15 minutes.

Best Hyperparameter Search Results.

```
▼ XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
               colsample_bylevel=0.9425384185492701,
               colsample_bynode=0.9095956806239844,
               colsample_bytree=0.706128679361455, device=None,
               early_stopping_rounds=None, enable_categorical=False,
               eval_metric=None, feature_types=None, gamma=1.6598135411398998,
               grow_policy=None, importance_type=None,
               interaction_constraints=None, learning_rate=0.40582798722369146,
               max_bin=None, max_cat_threshold=None, max_cat_to_onehot=None,
               max_delta_step=None, max_depth=7, max_leaves=None,
```

Different Scoring Metrics:

Accuracy

```
[170] opt.best_score_
```

```
0.46047061868056716
```

Recall Macro



```
opt.best_score_
```



```
0.2525865947425464
```

F1 Macro



```
opt.best_score_
```



```
0.26975596913050787
```

Precision Macro



```
opt.best_score_
```



```
0.41992192000607836
```

The accuracy scoring metric gave the best result.